KERNEL MONITOR

Linux System Resource Monitoring and Alert System

TEAM: TechNova

CPU
RAM
DISK
PROCESSES
LOAD

TEAM MEMBERS

Meet the Team

M. Subhash

Registration No.

2520030391

D. Namith

Registration No.

2520030249

E. Yeshwanth

Registration No.

2520030539

Abstract

Kernel Monitor is a lightweight C application for Ubuntu that reads Linux system information and displays resource usage in a single terminal dashboard. It refreshes every second and warns when CPU, RAM, or disk usage exceeds 80%.

MONITOR

CPU, RAM, disk, processes and load

INTERFACE

/proc files and statvfs()

OUTPUT

Live terminal readings and alerts

Problem Statement

Scattered system data

Users check several Linux commands or files to inspect system health.

Hard to interpret

Raw kernel counters are not immediately understandable to beginners.

Need for quick alerts

A simple dashboard can highlight resource usage above a chosen threshold.

KEY FUNCTIONALITY

Proposed Solution

CPU usage

Calculated from changes in /proc/stat counters

RAM usage

Uses MemTotal and MemAvailable in /proc/meminfo

Disk usage

Uses statvfs() for root filesystem capacity

Process count

Counts numeric PID directories in /proc

Load average

Reads 1-minute average from /proc/loadavg

Alerts

Warns above 80% CPU, RAM or disk

System Workflow

01

Start program

02

Read Linux resource interfaces

03

Calculate usage percentages

04

Display terminal dashboard

05

Check >80% thresholds

06

Wait 1 second and repeat

Tools and Technologies

Ubuntu Linux

Development and execution platform

C language

Reads and processes system statistics

GCC

Compiles KernelMonitor.c

Linux /proc

CPU, RAM, process and load statistics

POSIX / statvfs

Filesystem usage measurement

Terminal / gedit

Run and edit source code

Execution and Demonstration

```
gcc KernelMonitor.c -o KernelMonitor

./KernelMonitor          # LIVE: actual Ubuntu readings
./KernelMonitor --demo   # DEMO: simulated changing readings
```

LIVE readings may remain constant when system activity is stable. DEMO values are simulated and must not be presented as real measurements.

Expected Outcomes & Conclusion

Unified monitoring

One screen for five Linux system metrics

Automatic alerts

Threshold warnings when utilization exceeds 80%

Operating systems learning

Practical use of /proc, process IDs and file I/O

Future scope: configurable thresholds, historical logs, graphs and notifications.

Thank You

KERNEL MONITOR

TechNova | M. Subhash • D. Namith • E. Yeshwanth

Operating Systems & Systems Programming